

EARL LOUIS

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EDUCATION

University of Connecticut

Bachelor of Science in Robotics Engineering (Electronics Track)

Storrs, CT

Expected December 2026

TECHNICAL SKILLS

Competencies: Control Panel Design, P&ID; Development, Sensor Network Integration, Embedded Systems Design, Power Distribution, Process Automation, SCADA/HMI, PLC Systems

Software & Tools: Ignition (Certified), Python (multi-threading, signal processing), C, C++, AutoCAD, KiCad, LTSpice, AVR Microchip Studio

Protocols: Modbus TCP, EtherNet/IP, ROS, I2C, SPI, UART, IoT

PROFESSIONAL EXPERIENCE

Barnes Aerospace

Control Systems Intern

Windsor, CT

June 2025 – Present

- Reviewed PLC I/O designs and developed P&ID diagrams to support control panel design and equipment integration
- Supported integration of CNC machines and legacy equipment into the plant IoT network via Modbus TCP and EtherNet/IP, scoping connectivity requirements and configuring Ignition tags to surface real-time machine data and power status on operator dashboards
- Designed and implemented automated monitoring and control loops across facility operations, including chemical process control, safety equipment tracking, spill detection, and environmental compliance systems
- Earned Ignition 8.1 Credential and configured HMI/SCADA displays providing operators real-time data visualization

Norwich Public Utilities

Electrical Engineering Intern

Norwich, CT

June 2024 – January 2025

- Assisted in design and implementation of power distribution systems, ensuring compliance with industry standards
- Performed transformer testing and prepared phasing diagrams for high-voltage equipment installations
- Conducted field inspections supporting maintenance planning and system upgrades

Outlier AI

Machine Learning Quality Assurance Tester

Remote

May 2024 – January 2025

- Evaluated Meta AI-generated responses across technical, conversational, and health-related content areas, grading accuracy, relevance, and adherence to safety guidelines
- Flagged and downgraded responses where the model overstepped into medical or mental health advice, contributing to improved model safety

Demers Exposition Services, Inc.

Stage Technician

Uncasville, CT

December 2022 – June 2024

- Assisted crew leads with load-in and load-out of large-scale arena events including concerts and sporting events, running audio and lighting cable, staging equipment, and operating chain hoists during rigging setup and teardown

FEATURED PROJECTS

Drone Health Monitoring System

Raspberry Pi 5, I2C/SPI/UART, FFT, 915 MHz Radio

- Joint senior design project. Fitted a quadcopter with current, vibration, and temperature sensors. An onboard Raspberry Pi performed FFT-based processing and transmitted data over 915 MHz radio to a ground station running a live dashboard with automated out-of-range alarms.

Two-Wheeled Path Planning Robot

Romi 32U4, Raspberry Pi 3, Python, C++

- Navigation robot with a custom wavefront path planner and overhead camera pipeline for real-time localization and orientation detection.

Sound Intensity Visualizer

AVR 128DB48, C (AVR Microchip Studio), LED Dot Matrix, OLED Display

- Audio visualizer displaying live sound waveforms on an LED dot matrix with dB readout on a secondary OLED screen.